

1. Consider a modification to the activity-selection problem in which each activity a_i has, in addition to a start and finish time, a value v_i . The objective is no longer to maximize the number of activities scheduled, but instead to maximize the total value of the activities scheduled. That is, we wish to choose a set A of compatible activities such that $\sum_{a_k \in A} v_k$ is maximized.

i	1	2	3	4	5	6	7
s_i	3	2	5	1	8	7	10
f_i	7	5	9	4	9	9	12
v_i	3	4	2	3	1	2	3

- Determine an optimal solution for the given table.
- Give a polynomial-time algorithm for this problem.

1. Algorithm

$dp[i]$: 活動 $1 \sim i$ 的最佳值

對完成時間按升序排列

$$dp[0] = 0$$

for $i = 1$ to n ($n = 7$)

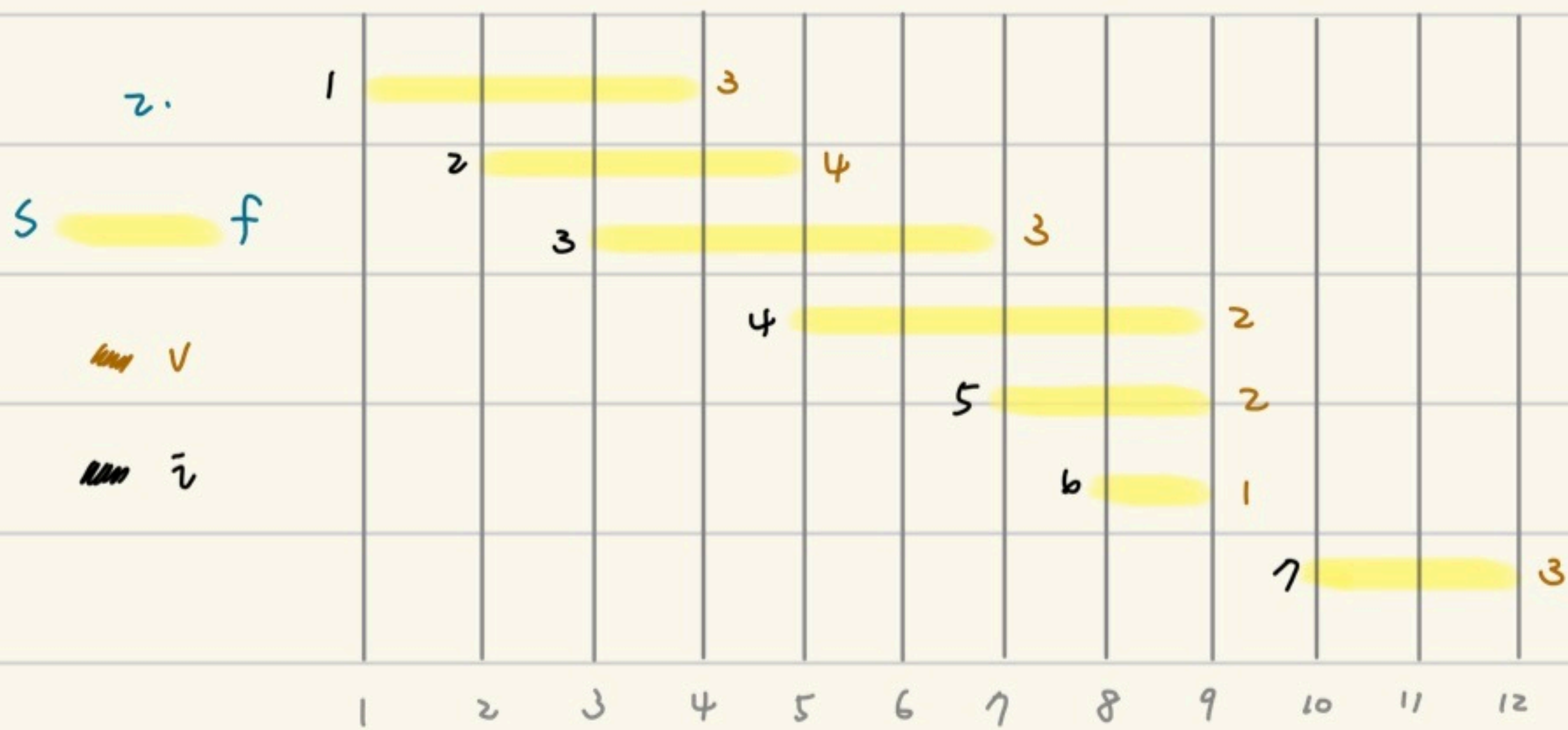
$last =$ 以二分搜尋法找完成時間 $\leq s_i$ 的活動 j $\rightarrow O(\lg n)$

$$dp[i] = \max \left(\underset{\substack{\downarrow \\ \text{目前活動不選}}}{dp[i-1]}, \underset{\substack{\downarrow \\ \text{選目前活動}}}{dp[last] + v_i} \right)$$

最後Ans為 $dp[n]$

$$\text{time complexity} = n \times O(\lg n) = O(n \lg n)$$

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dp 1 2 3 4 5 6 7

max (0, 0+3) (3, 0+4) (4, 0+3) (4, 4+2) (6, 4+2) (6, 4+1) (6, 6+3)

3 4 4 6 6 6 9